

No Variety Specification: $U_{ijt} = \beta_i X_{ijt} + \varepsilon_{ijt}$

In this specification, love of variety does not enter. Only preference for characteristics. We see that $\approx 95\%$ of the probability mass is on products 4, 5, and 6, with an inclusive value of 11.36. All that happened with the introduction of this product is mass shifted from $5 \rightarrow 6$. Product 6 = product 5 in value, thus it makes sense that it appears probability is being taken from 5.

Variety Specification: $U_{ijt} = \beta_i X_{ijt} + \gamma_i \log(1 + \Sigma^2) + \varepsilon_{ijt}$

where $\Sigma^2 = X_{jt} - \frac{1}{s} \sum_{s=1}^{t-1} X_{js}$ and $\gamma_i \in \{6, 9, 12\}$, we find that higher levels of γ lead to higher inclusive values. Interestingly, the addition of a product leads to stable choices of that product across the three variety levels, with lower γ more likely to select it due to its value being the same as 5's. IV decreases for all variety lovers. Increases in base model. Decreases are slight, though.

As the new product's quality changes, we see different trends. At $X_6 = 1.0$, the inclusive value changes are very close to at $X_6 = 5.0$. This is understandable, as we have fluctuation between X_1 and X_5 . But, when $X_6 = 3$, we see greater losses in inclusive value as no one picks X_3 anyways. Only the low γ type picks the new entrant at 0.21%.

High Quality Entrant:

No Variety (New Product, No Entry)	Product	CCP	Product	CCP
	Product 1	0.0012	Product 1	0.0016
	Product 2	0.0066	Product 2	0.0084
	Product 3	0.0332	Product 3	0.0437
	Product 4	0.1484	Product 4	0.1934
	Product 5	0.5927	Product 5	0.7529
	Product 6	0.2179	Product 5	0.7529
	% Δ Inclusive Value	1.0362	Inclusive Value	10.9606
$\gamma_i = 6$ (New Product, No Entry)	Product	CCP	Product	CCP
	Product 1	0.2973	Product 1	0.3313
	Product 2	0.0523	Product 2	0.0551
	Product 3	0.0029	Product 3	0.0027
	Product 4	0.0017	Product 4	0.0023
	Product 5	0.4769	Product 5	0.6086
	Product 6	0.1689	Product 5	0.6086
	% Δ Inclusive Value	0.9975	Inclusive Value	17.4522
$\gamma_i = 9$ (New Product, No Entry)	Product	CCP	Product	CCP
	Product 1	0.3643	Product 1	0.3971
	Product 2	0.0088	Product 2	0.0081
	Product 3	0.0001	Product 3	0.0
	Product 4	0.0001	Product 4	0.0002
	Product 5	0.4628	Product 5	0.5946
	Product 6	0.164	Product 5	0.5946
	% Δ Inclusive Value	0.9932	Inclusive Value	22.4506
$\gamma_i = 12$ (New Product, No Entry)	Product	CCP	Product	CCP
	Product 1	0.3913	Product 1	0.4234
	Product 2	0.0011	Product 2	0.0009
	Product 3	0.0	Product 3	0.0
	Product 4	0.0	Product 4	0.0
	Product 5	0.4494	Product 5	0.5756
	Product 6	0.1582	Product 5	0.5756
	% Δ Inclusive Value	0.9906	Inclusive Value	27.5139

Low Quality Entrant:

No Variety (New Product, No Entry)	Product	CCP	Product	CCP
	Product 1	0.0016	Product 1	0.0016
	Product 2	0.0089	Product 2	0.0084
	Product 3	0.0439	Product 3	0.0437
	Product 4	0.1926	Product 4	0.1934
	Product 5	0.7522	Product 5	0.7529
	Product 6	0.0008	Product 5	0.7529
	$\% \Delta Inclusive Value$	0.9994	Inclusive Value	10.9606
$\gamma_i = 6$ (New Product, No Entry)	Product	CCP	Product	CCP
	Product 1	0.2665	Product 1	0.3313
	Product 2	0.0446	Product 2	0.0551
	Product 3	0.0025	Product 3	0.0027
	Product 4	0.0019	Product 4	0.0023
	Product 5	0.5796	Product 5	0.6086
	Product 6	0.1049	Product 5	0.6086
	$\% \Delta Inclusive Value$	0.9961	Inclusive Value	17.4522
$\gamma_i = 9$ (New Product, No Entry)	Product	CCP	Product	CCP
	Product 1	0.3131	Product 1	0.3971
	Product 2	0.0072	Product 2	0.0081
	Product 3	0.0001	Product 3	0.0
	Product 4	0.0001	Product 4	0.0002
	Product 5	0.5616	Product 5	0.5946
	Product 6	0.1179	Product 5	0.5946
	$\% \Delta Inclusive Value$	0.9932	Inclusive Value	22.4506
$\gamma_i = 12$ (New Product, No Entry)	Product	CCP	Product	CCP
	Product 1	0.3321	Product 1	0.4234
	Product 2	0.0009	Product 2	0.0009
	Product 3	0.0	Product 3	0.0
	Product 4	0.0	Product 4	0.0
	Product 5	0.5451	Product 5	0.5756
	Product 6	0.1219	Product 5	0.5756
	$\% \Delta Inclusive Value$	0.9908	Inclusive Value	27.5139

Medium Quality Entrant:

No Variety (New Product, No Entry)	Product	CCP	Product	CCP
	Product 1	0.0016	Product 1	0.0016
	Product 2	0.0087	Product 2	0.0084
	Product 3	0.0429	Product 3	0.0437
	Product 4	0.1883	Product 4	0.1934
	Product 5	0.7375	Product 5	0.7529
	Product 6	0.021	Product 5	0.7529
	% Δ Inclusive Value	1.0017	Inclusive Value	10.9606
$\gamma_i = 6$ (New Product, No Entry)	Product	CCP	Product	CCP
	Product 1	0.3295	Product 1	0.3313
	Product 2	0.0571	Product 2	0.0551
	Product 3	0.0031	Product 3	0.0027
	Product 4	0.0021	Product 4	0.0023
	Product 5	0.6069	Product 5	0.6086
	Product 6	0.0013	Product 5	0.6086
	% Δ Inclusive Value	0.9835	Inclusive Value	17.4522
$\gamma_i = 9$ (New Product, No Entry)	Product	CCP	Product	CCP
	Product 1	0.3966	Product 1	0.3971
	Product 2	0.0093	Product 2	0.0081
	Product 3	0.0001	Product 3	0.0
	Product 4	0.0001	Product 4	0.0002
	Product 5	0.594	Product 5	0.5946
	Product 6	0.0	Product 5	0.5946
	% Δ Inclusive Value	0.9823	Inclusive Value	22.4506
$\gamma_i = 12$ (New Product, No Entry)	Product	CCP	Product	CCP
	Product 1	0.4234	Product 1	0.4234
	Product 2	0.0011	Product 2	0.0009
	Product 3	0.0	Product 3	0.0
	Product 4	0.0	Product 4	0.0
	Product 5	0.5755	Product 5	0.5756
	Product 6	0.0	Product 5	0.5756
	% Δ Inclusive Value	0.9818	Inclusive Value	27.5139